

AI Talent Acquisition Platform ? Full Project Documentation

1. Project Overview

AI Talent Acquisition Platform is an enterprise-grade recruitment automation system that uses Arti

Key Capabilities

- * Automated resume parsing using NLP (Named Entity Recognition)
- * Intelligent candidate-job matching using BM25 + Vector Search
- * AI-powered interview question generation
- * Voice interview support with Whisper speech-to-text
- * Graph-based skill relationship analysis (Graph RAG)
- * Explainable AI recommendations
- * Real-time analytics dashboards
- * Production-ready with Docker, CI/CD, and monitoring

2. Technology Stack

Layer	Technology
Backend Framework	FastAPI (Python)
Database	PostgreSQL + SQLAlchemy ORM
Migrations	Alembic
Authentication	JWT (python-jose) + bcrypt
Vector Database	Qdrant (in-memory for dev, ser
Embeddings	sentence-transformers (all-Min
NLP	spaCy + NLTK
Lexical Search	BM25 (rank-bm25)
LLM	OpenAI / Azure OpenAI (GPT-4o-
Speech-to-Text	OpenAI Whisper
Graph Database	Neo4j (with rule-based fallback
Email	SMTP + Jinja2 templates
Containerization	Docker + docker-compose
CI/CD	GitHub Actions
Deployment	Azure Container Apps
Monitoring	Custom middleware + drift dete

3. Architecture Diagram



? SQL ? ?(Vectors) ? ? (Graph) ? ? APIs ?
????????? ?????????????? ????????????? ?????????????

4. Database Schema (PostgreSQL)

Entity Relationship

Users (id, email, hashed_password, full_name, role, is_active, phone)
?
??? Candidates (id, user_id, headline, skills[], experience_years, education[])
? ?
? ??? Resumes (id, candidate_id, file_path, status, extracted_skills[], parsed_data)
? ?
? ??? JobApplications (id, job_id, candidate_id, status, match_score)
? ?
? ??? Interviews (id, application_id, type, status, questions[], responses[])
? ?
? ??? Evaluations (id, interview_id, overall_score, strengths[], recommendation)
?
??? Jobs (id, title, description, required_skills[], status, created_by)

Models

- * Users ? HR, Candidate, Admin roles with RBAC
- * Candidates ? Profile with skills, experience, education (JSON fields)
- * Jobs ? Job descriptions with required/preferred skills
- * JobApplications ? Links candidates to jobs with match scores
- * Resumes ? File metadata + parsed NLP data
- * Interviews ? AI-generated questions + candidate responses
- * Evaluations ? Multi-dimensional scoring with explainable recommendations

5. API Endpoints (62 Total)

5.1 Authentication (4 endpoints)

Method	Endpoint	Description
POST	/api/v1/auth/register	Register new user (HR/Candidat
POST	/api/v1/auth/login	Login and receive JWT tokens
POST	/api/v1/auth/refresh	Refresh access token
GET	/api/v1/auth/me	Get current user profile

5.2 Users (5 endpoints)

Method	Endpoint	Description
GET	/api/v1/users/	List users (HR/Admin only)
GET	/api/v1/users/{id}	Get specific user
PUT	/api/v1/users/me	Update profile
POST	/api/v1/users/me/change-passwo	Change password
DELETE	/api/v1/users/{id}	Deactivate user (Admin)

5.3 Jobs (7 endpoints)

Method	Endpoint	Description
POST	/api/v1/jobs/	Create job description
GET	/api/v1/jobs/	List jobs (filtered by role)
GET	/api/v1/jobs/{id}	Get job details
PUT	/api/v1/jobs/{id}	Update job
DELETE	/api/v1/jobs/{id}	Delete job
POST	/api/v1/jobs/{id}/apply	Candidate applies to job
GET	/api/v1/jobs/{id}/applications	List applications

5.4 Resumes (6 endpoints)

Method	Endpoint	Description
POST	/api/v1/resumes/upload	Upload resume (PDF/DOCX/TXT)
POST	/api/v1/resumes/{id}/parse	Parse resume with NLP
GET	/api/v1/resumes/	List resumes
GET	/api/v1/resumes/{id}	Get resume details
POST	/api/v1/resumes/match-jd	Match candidates to JD
POST	/api/v1/resumes/search	BM25 lexical search

5.5 Search & Ranking (5 endpoints)

Method	Endpoint	Description
POST	/api/v1/search/semantic	Vector semantic search
POST	/api/v1/search/hybrid	BM25 + Vector hybrid search
POST	/api/v1/search/index-resume	Index resume in vector DB
POST	/api/v1/search/index-all	Batch index all resumes
GET	/api/v1/search/vector-stats	Vector DB statistics

5.6 Notifications (3 endpoints)

Method	Endpoint	Description
POST	/api/v1/notifications/shortlis	Send shortlist email
POST	/api/v1/notifications/intervie	Send interview invite
POST	/api/v1/notifications/bulk-not	Bulk notifications

5.7 Analytics (5 endpoints)

Method	Endpoint	Description
GET	/api/v1/analytics/overview	Dashboard metrics
GET	/api/v1/analytics/job-pipeline	Job hiring pipeline
GET	/api/v1/analytics/skills	Skill distribution
GET	/api/v1/analytics/experience	Experience distribution
GET	/api/v1/analytics/funnel	Hiring funnel

5.8 Interviews & Evaluation (6 endpoints)

Method	Endpoint	Description
POST	/api/v1/interviews/generate-qu	AI question generation
POST	/api/v1/interviews/submit-resp	Submit answers
POST	/api/v1/interviews/evaluate	AI evaluation
POST	/api/v1/interviews/rerank	Re-rank after interview
GET	/api/v1/interviews/{id}	Get interview details
GET	/api/v1/interviews/by-applicat	List interviews

5.9 Voice Interview (4 endpoints)

Method	Endpoint	Description
POST	/api/v1/voice/create-session	Start voice session
POST	/api/v1/voice/upload-response	Upload audio answer
POST	/api/v1/voice/transcribe/{id}	Whisper transcription
GET	/api/v1/voice/status	Service status

5.10 Knowledge Graph & Graph RAG (9 endpoints)

Method	Endpoint	Description
GET	/api/v1/graph/status	Graph DB status
GET	/api/v1/graph/related-skills/{	Find related skills
POST	/api/v1/graph/add-skill	Add skill to graph
POST	/api/v1/graph/add-relationship	Create skill relationship
POST	/api/v1/graph/index-candidate/	Index candidate
POST	/api/v1/graph/index-job/{id}	Index job
POST	/api/v1/graph/skill-gap	Skill gap analysis
POST	/api/v1/graph/evaluate	Graph RAG evaluation

POST	/api/v1/graph/recommendations	Learning path
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5.11 Operations & Monitoring (6 endpoints)

Method	Endpoint	Description
GET	/api/v1/ops/health	Detailed health check
GET	/api/v1/ops/metrics	Latency & request metrics
GET	/api/v1/ops/drift	Model drift detection
POST	/api/v1/ops/check-prompt	Prompt injection check
GET	/api/v1/ops/costs	API cost summary
GET	/api/v1/ops/audit-logs	Audit trail

6. AI/ML Components

6.1 Resume Parsing (NLP)

- * Tokenization ? NLTK word_tokenize
- * Lemmatization ? NLTK WordNetLemmatizer
- * Stop-word Removal ? NLTK English stopwords
- * Named Entity Recognition ? spaCy en_core_web_sm
- * Extraction: Name, Email, Phone, Skills, Experience, Education

6.2 Search & Ranking

- * BM25 (Lexical) ? Term frequency-based ranking
- * Vector Search (Semantic) ? 384-dim embeddings, cosine similarity
- * Hybrid Search ? Configurable weights (BM25 35% + Vector 45% + Skills 20%)
- * ANN/HNSW ? Qdrant's internal indexing for fast retrieval

6.3 AI Interview System

- * Question Generation ? Prompt engineering with context augmentation
- * Difficulty Adjustment ? easy/medium/hard/mixed
- * Categories ? Technical, Behavioral, Situational, Problem-solving
- * Evaluation ? Multi-dimensional (technical, communication, problem-solving)
- * Faithfulness Scoring ? Detects vague/overconfident responses
- * Explainable Recommendations ? hire/maybe/reject with justification

6.4 Graph RAG

- * Knowledge Graph ? Skills, relationships, hierarchies
- * Graph Traversal ? Find related skills within N hops
- * Skill Gap Analysis ? Matched vs missing vs transferable skills
- * Graph-Enhanced Evaluation ? Context-enriched recommendations
- * Learning Path ? Identifies skills easy to learn based on existing knowledge

6.5 Voice Interview

- * Audio Upload ? MP3, WAV, WebM, OGG, M4A
- * Whisper STT ? OpenAI Whisper API for transcription
- * Segment-level Timing ? Per-sentence timestamps
- * Auto-evaluation ? Transcripts fed to AI evaluator

7. Security & Governance

- * JWT Authentication ? Access + Refresh tokens
- * RBAC ? Admin, HR, Candidate roles with endpoint-level guards

- * Password Hashing ? bcrypt
- * Prompt Injection Protection ? Pattern detection + sanitization
- * Cost Monitoring ? Per-model token usage tracking
- * Audit Logging ? All actions recorded with timestamps
- * Input Validation ? Pydantic schemas for all requests

8. DevOps & Deployment

Docker

```
docker-compose up    # PostgreSQL + App (auto-migration)
```

CI/CD Pipeline (GitHub Actions)

- 1. Lint ? ruff + mypy
- 2. Test ? pytest with PostgreSQL service
- 3. Build ? Docker image ? GitHub Container Registry
- 4. Deploy ? Azure Container Apps

Environment Variables

```
DATABASE_URL=postgresql://user:pass@host:5432/db
SECRET_KEY=your-secret-key
OPENAI_API_KEY=sk-...           # Optional: enables LLM features
QDRANT_HOST=localhost           # Optional: defaults to in-memory
NEO4J_URI=bolt://localhost:7687 # Optional: falls back to rule-based
SMTP_HOST=smtp.gmail.com       # Optional: enables email
```

9. Setup Instructions

Local Development

```
cd ai-talent-platform
python -m venv venv
.\venv\Scripts\activate      # Windows
pip install -r requirements.txt
python -m spacy download en_core_web_sm
copy .env.example .env       # Edit with your settings
uvicorn app.main:app --reload --port 8000
```

With Docker (Recommended)

```
docker-compose up --build
# App: http://localhost:8000
# Docs: http://localhost:8000/docs
```

Run Tests

```
pytest tests/ -v
```

10. Sprint Roadmap Summary

Sprint	Week	Goal	Story Points	Endpoints
1	Week 1	Foundation (Auth + DB + APIs)	26	16
2	Week 2	Resume Intelligence (NLP + BM25)	26	6
3	Week 3	Semantic Search & Ranking	34	13
4	Week 4	GenAI Evaluation	26	6
5	Week 5	Voice Interview + Graph RAG	42	13
6	Week 6	MLOps & Deployment	60	8
Total			~214	62

11. Project Structure

```
ai-talent-platform/
??? app/
?   ??? main.py                # FastAPI entry point
?   ??? core/
?   ?   ??? config.py          # Settings (env-based)
?   ?   ??? security.py        # JWT + password hashing
?   ?   ??? dependencies.py     # Auth + RBAC dependencies
?   ?   ??? middleware.py      # Latency tracking
?   ??? api/v1/
?   ?   ??? router.py          # Route aggregator
?   ?   ??? endpoints/
?   ?       ??? auth.py        # Login, register
?   ?       ??? users.py       # User management
?   ?       ??? jobs.py        # Job CRUD + apply
?   ?       ??? resumes.py     # Upload, parse, match
?   ?       ??? search.py      # Semantic + hybrid search
?   ?       ??? notifications.py # Email automation
?   ?       ??? analytics.py   # Dashboard data
?   ?       ??? interviews.py  # AI questions + evaluation
?   ?       ??? voice.py       # Voice interview
?   ?       ??? graph.py       # Knowledge graph + RAG
?   ?       ??? ops.py         # Monitoring + governance
?   ??? models/                # SQLAlchemy ORM (6 models)
?   ??? schemas/               # Pydantic validation
?   ??? services/              # Business logic (10 services)
?   ?   ??? resume_parser.py   # NLP extraction
?   ?   ??? jd_matcher.py      # BM25 matching
?   ?   ??? embedding_service.py # Vector embeddings
?   ?   ??? vector_store.py    # Qdrant operations
?   ?   ??? hybrid_search.py   # Combined search
?   ?   ??? llm_service.py     # OpenAI/Azure wrapper
?   ?   ??? question_generator.py # AI question gen
?   ?   ??? evaluation_service.py # AI evaluation
?   ?   ??? reranking_service.py # Post-interview ranking
?   ?   ??? voice_service.py   # Whisper STT
?   ?   ??? knowledge_graph.py # Neo4j graph
?   ?   ??? graph_rag.py       # Graph RAG
?   ?   ??? email_service.py   # SMTP notifications
?   ?   ??? analytics_service.py # Metrics
?   ?   ??? monitoring.py      # Observability
?   ?   ??? governance.py      # Security + audit
?   ??? db/                    # Database session + base
??? alembic/                   # Database migrations
??? tests/                     # Test suite
??? scripts/                   # Utility scripts
??? uploads/                   # File storage
??? .github/workflows/ci-cd.yml # GitHub Actions pipeline
??? Dockerfile                 # Container build
??? docker-compose.yml         # Full stack orchestration
??? requirements.txt            # Python dependencies
??? .env.example               # Environment template
```

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Repository: https://github.com/NAVEEN1901/AI_INTERVIEWER_SYSTEM_DESIGN